GHamine

Systems engineer

GitHub: github.com/GHamine / Site: ghamine.dev

SUMMARY

I build full-stack systems and measure whether their stated invariants hold. The three case studies on this ship the fixture, the artifact path and the limits behind every number.
site

SELECTED SYSTEMS

Ledger Lock

PostgreSQL, Node.js, pgTAP

- Held 1 unit of stock against 20 concurrent buyers: 1 committed reservation, 19 rejection events, 0 oversold units across 3 seeded runs.
- Moved the rejection insert inside the reservation transaction after commit 4f1c2ab left 2 outcomes for 1 decision; the seed now runs a forced-rollback case before each release.

Merge Gate

GitHub Actions, TypeScript, SQLite

- Blocked 1 of 1 red pull requests and let 0 red merges reach main by making policy-gate a required check backed by rules stored in SQLite.
- Fetches the head SHA at check time after commit b7d9e10 judged the previous SHA on a force-push; the fixture now includes that case.

Citation Desk

Python, FastAPI, SQLite

- Returned 12 of 12 answers with a resolvable passage id on the 12-question fixture, with 0 uncited answers shipped and 3 explicit refusals.
- Replaced embeddings with FTS5 retrieval: 1 correct answer lost, 400 MB model download and 3 s cold start removed.

SKILLS

PostgreSQL / SQLite / FTS5 / Node.js / TypeScript / Python / FastAPI / GitHub Actions / pgTAP / Astro

VERIFICATION

- npm run verify runs astro check, tsc, the asset and content contracts, the copy and accessibility audits, the JS budget, and a static build.
- Every numeric claim links to a committed artifact under public/assets/<slug>/.
- The site ships no client-side JavaScript and no tracking.

LIMITS

- The fixtures are seeded and small: 1 unit, 1 pull request, 12 questions.
- The measurements prove the stated invariants hold on those fixtures, not that they hold at production scale.
- The repositories are snapshots; the case pages do not call live services.

Generated 2026-10-23 from src/data/resume.json